Basics of Active Inference (Discussion) ~ Ben White ~ Active Inference for

Courses/ActiveInferenceForTheSocialSciences | Ben White ~ Active Inference for the Social Sciences 2023 | 1:33:17

all right July 25th 2023
and we are in the discussion section
basics of active inference this is the
first discussion section that we've had
for this course so thank you all for
joining and it's going to be regarding
the topics that Ben White introduced in
his recent lecture so everyone will be
welcome to pop in and introduce
themselves and then I know that Ben has
some ideas to discuss and many other
spontaneous and written questions will
come into play so
I will uh with that exit for now and
pass to perhaps some of our first time
live stream guests
I guess I've already spoken so uh I may
as well continue so I'm Darius
um I am Master student I'm just about to
finish at UCL
and sort of social distributed cognition
I work in the social cognition lab
looking at salience regulation and
attentional mechanisms within social
contexts and but all within an active
inference framework
um within the Bayesian brain hypothesis
framework and yeah it's a sort of recent
Discovery and Obsession and so I'm sort
of going to grips with the both the high
road and the low road
um and so I've had the opportunity to
chat to Mark and some other researchers
who have been super informative but
always looking to learn more and get my

head around even more of the sort of
philosophical and mathematical Theory
oh
sounds good
so hey everyone can you hear me
yep
I'm Francisco balzan I'm a PhD student
at the University of Bologna Italy
actually really working on the
intersection between artificial
intelligence and education and I have a
background in cognitive anthropology and
philosophy of science and
I'm at excellent I'm actually in a
a few years ago
so I stepped out into the rabbit program
active inference and currently pretty
much interested in uh
um multi-scale active inference models
are the scientific technicians so
pretty interested about the agent level
modeling of scientific reasoning the
assumption that might emerge from the
interaction with the scientific
environment so we're referring to
Scientific music construction and all
this type of
top-down environment interactions so
super interested to
do something from you thank you
thank you very much
would anybody else like to introduce
themselves before we start
thank you
so you can first okay hi hi I'm Regina
Sagi umina I'm from Guatemala and well
I'm not from the social sciences I come
from biology and Neuroscience and um I'm

here because
um I'm working on a project as a
technical operator in this big project
in DX Escape material mines and I do the
technical part and experiments but I
want to learn the philosophical part
that's the background that you know
makes The Theory of Everything we're
going to you know experiment so I'm here
to learn
um and also I am like you know when you
see the Olympics that you you see all
the swimmers and you always say I wish
there was a normal guy to see how good
these people are so you could compare so
that's me today right I I understand
half of what's going on but I'm happy to
be here
so yeah hi
oh yeah sorry I'm late um sorry my
name's Lee um I'm a PhD student at the
uh University of New York
um studying systems transformation
um and I'm I'm joining because I'm using
uh perceptual control theory at the
minute to uh to model examples of
effective practice
um in organizations and and I've read
quite a lot of um
active inference papers and I understand
that there's a lot of resonance uh and
overlap between perceptual control
theory and
um active influence although I
understand it's within a predicted
framework and also it's uh it's a kind
of a level of abstraction higher so
you're able to kind of quantify the

difference between
um sort of the the predictive State or
the outcome State and you know in the
current state so what I'm really trying
to understand is how might that level of
abstraction be useful
um in what I'm doing actually because I
understand it's a you know a much more
kind of concurrent Theory and framework
than um then perceptual control theory
yeah very cool
um anybody else
I think there's was that everybody
okay well um was everybody
um present at the has everybody seen the
lecture that I gave a couple of weeks
ago
okay so what I think it might be helpful
then if I kind of very very briefly go
over what what we covered in that
lecture just to kind of jog people's
memories and then we can maybe pick up
some questions from there
um so the lecture last week was on the
basics of active inference and it was
um
intended to be an introduction to the
framework on a very abstract
philosophical level so it left out all
of the kind of mathematics and Technical
aspects of the framework and the
objective really was to lay the
groundwork for future weeks in this
course because
um this is obviously a course on active
inference in the social sciences and so
in the subsequent weeks we're going to
be looking at topics like Collective

Behavior social cognition uh shared
norms and Niche construction and so what
I wanted to do was to put on the table
um
a kind of
fully fleshed out picture of what the
individual agent looks like in active
inference and so to do this
um we looked at uh emotion agency uh
mind so we looked at kind of the role of
action
um and perception
um the role of internal representations
um inactive imprints and then as a case
study we kind of we looked at a active
inference based account of addiction to
kind of bring all of these threads
together
um to kind of articulate a lot of the
things that that I'd said in the
previous sections
um
so
I'm I mean the questions here don't have
to be specifically
um Tethered to that lecture I suppose we
can just start with any general
questions about active inference and
philosophy and how active inference
applies to individual agents or if
anybody did want to pick up on any
threads from the lecture then we can go
from there
can I ask something just for the basic
questions yeah of course
um I don't understand because I think
you explain it later but how
how does creativity and

science in general enters active
imprints
how does trying to be creative and not
put doing what you predict enters this
like role yeah yeah this is a really
good question and I think um so were you
there have you did you see the lecture
yeah but I remember half of it yeah
that's that's fine but I think uh I
think do you remember the dark room
problem yeah yeah I know that's that's
where the answer is but yeah well that's
it's it's it's it's I think there are
several
interested in that dark room problem
some of them are some of them more
interesting than others and I I tried to
cover I did obviously didn't have time
to cover all of those in the lecture
um but the kind of for those that
perhaps aren't familiar the dark room
problem is um this kind of canonical
philosophical worry
um about predictive processing and
active inference that says essentially
if it's if all we're trying to do is
minimize prediction errors why don't we
um just find maximally predictable
environments and you know why do we not
just lay in a dark room hooked up to an
intravenous trip and enjoy a kind of
error-free Lifestyle
um so I think that's the kind of that's
the question that you're articulating
there
um I just quickly see a hand from Darius
Darius did you want to jump in or is it
a separate question

it's a separate question so I would just
do it in advance but yeah no ignore me
I'll Circle back around in a second then
you're free um I should say actually
um this is the first time I've kind of
shared a discussion like this so I'll be
explicit and say um people should just
feel free to jump in whenever they want
if you want to take the Disco If you
want to kind of um add something or or
kind of build on a question that we're
answering feel free to jump in
um but yeah with it with a dark room
problem the as I as I'm aware the first
answer to that question and this is
something that I did Cover in the
lecture is
essentially a big part of active
inference is to recognize that the
agents are it's a kind of fundamentally
embodied framework okay so what that
that can be kind of cashed out in
various ways but one of the one of the
most important ways is that the the
expected states of the organism or that
the agent are necessarily rooted in the
phenotype of the organism and the the
kind of evolutionary biological needs
that come with having a particular body
so there was a paper that I referenced
in a lecture by
that's three authors on this paper it's
Andy Clark call for assistant and then
I'm forgetting the name of the third
author if anybody can recall
um feel free to drop it in the chat but
this is this is an answer to the dark
room where it builds on that notion of

embodiment and says look
um creatures like us humanism
in the dark room and do nothing and just
uh kind of enjoy the very predictable
March of hunger Thursday Etc but
obviously those particular biology of
the room and so you have this very
fundamental level
um
we have this very fundamental um kind of
basis upon which we have these needs
that need to be met and so we need to
engage in exploratory behaviors
um on the topic of novelty
um because that's because that's the
kind of Base I take that to be a kind of
Baseline answer
um but there are clearly
um kinds of activities and behaviors
that we do engage in
um going beyond the dark room kind of
curiosity play
um creativity engaging in kind of why do
we create works of art and it there's
not an obvious link between those kinds
of behaviors and the kinds of biological
needs
um that are emphasized in the original
answer to the dark room worry
um so in answer to Regina's question
about creativity I think that a good
starting point for this is the section
on aerodynamics that I introduced in the
lecture so this was the idea that it
there's this strong connection between
affectivity as a kind of emotional
embodied feeling State and the changes
to the rate in prediction error

minimization so I don't know
um I am now very much uh articulating
Mark's Work here so I don't know if Mark
wants to jump in at any point and do a
much better job than I can
um
yeah I think there's there's certainly
there's certainly work on the horizon
well I know that there are people who
are currently building a series of
artistic pretty that are
that are based on this this idea of
aerodynamics so Mark's paper on play
with Mark andon that I mentioned has
really interesting idea where if you
explain
playful behaviors which are kind of
fundamentally created and very we are
you know it's inherently creative
exercise I think the really interesting
idea that comes out of that is one thing
that we engage in because it feels good
is we we create niches of very
manageable prediction error that we can
then minimize
um so yeah that's playful behavior on is
the kind of the answer to the puzzle of
play so why do we in the puzzle of the
players why do we engage in previous uh
metabolically quite costly
um
and they don't have any obvious benefit
and the idea there's that well there's
there's kind of several ideas in there
um but one of them is that it just
literally feels good because minimizing
prediction error feels good to us so
whenever we do better than expect at

minimizing prediction error it feels
good and so I think so obviously I don't
know for sure but I think that any
answer that we give regarding artistic
creativity is going to fall within that
kind of bulk that essentially we engage
um in the part of the creative process
is just constructing those that give us
tasks
games of manageable prediction error
that ideally sit at the boundaries of
our skill capability can I add two quick
things there Ben so just so that was a
great explanation but just to hit two
points a little bit harder one
you might think
um all that matters existing error
that's why it looked a little bit
paradoxical that we also pre
antithetical to our modus operandi which
is to reduce error but when you remember
that for our contradiction error
minimizing system we have a really deep
temporal model we're managing
uncertainty at a big Horizon uh maybe
even multi-generational you know where
we're thinking about our kids or our
kids I mean who knows how deep the
generative model actually runs it turns
out that
that stops developing error minimizing
skills and abilities and just hangs out
in one Micro Niche because that Micro
Niche is going to be upset sooner or a
little bit right I mean it's such a
great example of where we thought we
were in a really set Vector State and
then suddenly it gets jostle then all of

us go oh my goodness like what do we do
we've been bumped
um and uh it turns out that the best air
minimize will be the
responsive to kinds of errors that are
Advanced and it hates those errors are
digestible so that means not too complex
that you can't do anything with it you
can't learn anything from it but also
not so boring that there's nothing to
learn if we hang out if we're sensitive
to and we hang out at the edge of our
capability then we keep developing new
error minimizing abilities which
actually sets us up to be good error
minimizing systems over the long run so
even though we're investing metabolism
now we're getting we're setting
ourselves up to be able to manage you
know especially in the deep end of the
pool Black Swan funds manage uncertainty
that we're not going to be able to
predict the really unpredictable
unpredictable
um that comes from hanging out at this
Edge so part of our curiosity and
playfulness and creativity are going to
be about us making and digesting novel
slopes of volatility I think that's one
that's one really great answer for why
you get playfulness and curiosity out of
this and creativity I'll just drop one
more and we don't have to get into it
too deeply here but here's one other one
that we're thinking of
there's something really special in the
creation of art especially in a public
sphere that I think is so interesting

and that somebody needs to be sort of
looking at this
um we there's actually a new collection
coming out in philosophical Trends B
um on Art and predictive processing
which I think you should check out if
you're interested in these topics
um but the idea that we've been thinking
about is there are ways that we can
bring our generative model out and put
it into a public sphere you can take
something which is typically on the
inside and you can put it outside
and then you can have other error
minimizing systems look at it and Fiddle
around with it and then we can re-imbibe
it
um I mean we're doing this all the time
I mean that's what language allows us to
do and what writing allows us to do
think about maths you're bringing a part
of your predictive understanding you're
laying it out and other predictive
agents can fiddle around with it and
then you can take it back in as part of
your updating your own model and I don't
know how much more I can say here I just
think this is a kind of horizon but I
think there's a really juicy thing to
say here about where real art is like
Tolstoy maybe Leo tostai was already on
this where you put something of the
Creator in the creation and then other
people receive that and then they can
they can um they can do something else
with it and then you're able to sort of
take it back in all under the guise of
sort of minimizing volatility or

understanding volatility
maybe that was a bit deep but uh no that
was really that was pretty good yeah
yeah
um I just didn't so in the chat I just
dropped a link to um
um yeah so somebody just asked about
papers related to hanging out at the
edge of our capabilities I'll do that
right now uh yeah sure awesome papers on
that uh Mark's got a few I already
dropped a link in there to an aeon
article by um some of the old expect
project crew including Mark Kate Nave
George Dean and Andy Clark on the value
of uncertainty which is
I think it's a really good start point
for some of these questions
um just because it's aeon is a kind of
non-academic
um place where you can you know it's a
it's a non-technical introduction and
there's a wonderful example in that
paper of a guy named I think it's Max
Hawkins is that right Mark Max who um
he's a kind of tech guy and he realized
he was getting kind of bored with his
life he inadvertently trapped himself in
a dark room because his life had become
so
wrote and scheduled and systematic he
realized that he would be kind of
incredibly easy to kidnap because he was
in the same place at the same time every
single day and to get he took drastic
action to break out of that dark room so
he he kind of introduced a
randomization algorithm that would it

would kind of choose for him where he
was going to eat where he was going to
go which shows he was going to attend
and this example is a really nice
centerpiece in this article so and I
think yeah I think this is a nice place
to start because it's also grounded in
discussions about epistemic actions as
well because there's also you know as
cool as play and art and creativity are
there's also really really good reasons
that uncertainty is valuable as well
right because
while we might want to exploit all of
the opportunities for kind of
nourishment and valuable predictionary
minimization in our environment there
are of course going to be times where
um those those opportunities are
exhausted and we need to go and explore
different ones and I think there's some
there's there's some stuff in there as
well on epistemic actions within the
context of navigation as well so
sometimes we're going to have to we're
going to have to temporarily move
further away from our Target in order to
minimize our uncertainty about where
we're going so there's lots and lots of
of stuff on the value of uncertainty
um and it kind of it's very much at the
Forefront of some of the really
interesting philosophical applications
of the framework for sure
okay yeah by all means yeah we've got
time you just said something I know you
have a question that is but
with the value of uncertainty it's just

that I'm thinking because I've been
working a lot with The Magicians and
there's a way of using uncertainty as
entertainment and it's very difficult to
understand it's like you know as a
concept for entertainment why would you
like to be in a place where you cannot
you want to predict but you cannot and
you still want to participate in this
activity so for me magic shows it's
gonna break in a little bit
um
yeah I think my I spoke to um
I forget the person's name but there was
a Escape meeting here at Sussex and I
spoke to one of the guys on Xscape who
had been working on Magic
um I don't know if you know who I'm
talking about but I think they have a
book on on Magic
um and I think that's really cool to
think about from a predictive process in
perspective the idea that magic is just
these wonderful kind of violations of
expectations in a certain sense I think
predictive processing provides a nice
really intuitive framework for thinking
about those kinds of things one one
little point there uh Regina just about
again if it feels paradoxical just to
sort of um complexify your view remember
that the this is always happening in a
hierarchical system
so just because you have errors at one
level of the system doesn't mean you're
going to have sort of critical cascading
unbearable errors at higher levels so
that's why it's fun to go to horror

movies so I dropped the paper here also
we have a new paper coming out on on
horror movies I'm pretty interested in
this
um but it's the same as the it's the
same as going to the magic show in some
way now the reason why those errors are
are fun is because we're safe in a
theater
um we're with our friends we have lots
of sugar in our system from eating
popcorn and Candy
um we can control the amount of scary by
by you know covering our eyes or like
there's lots of control here and yet
we're getting media that is that's
pinging all of our evolutionarily
ancient error tracking systems so that
we're getting jumps in the physiology as
if there's a bunch of volatility that we
need to manage and yet we're in a
completely safe space which is really
fun for our kind of system because we're
hierarchy LED predictive systems the
high level here isn't jeopardized it
knows I'm in a theater but the low level
stuff is still registering all sorts of
little volatilities but of course they
get squashed they don't Cascade all the
way up
um or or if you're the kind of person
where they do tend to cast all the way
up then you're also the kind of person
who doesn't like going to horror movies
because it kind of scared like you go
home and think oh goodness maybe ghosts
are real and maybe I live with them you
wouldn't be going there then so with

magic it's the same sort of thing
um we know to a certain against so
with errors
I don't know if you've ever seen um
who does the really extreme magic David
where he like was in a block yeah David
Blaine have you ever seen him he goes to
Haiti where you have a community that
really believes in Magic and he does
some magic and he gets into trouble like
suddenly all the young men are like oh
no no no no that's that's no bueno that
that you're doing this and then he had
to the camera pans back and he goes no
no hold on I'll show you here I'll show
you how it's done it was just a trick
I'll just show you outside it's not real
magic because there they're not having
fun with it anymore they're like no
that's like not a good thing to do you
know so it just goes to show that like
um
our our engagement with magic is mostly
a sort of scary play yeah
I think one thing I just kind of sorry
about the seagulls I don't know if you
can hear those there are seagulls going
crazy just outside my window
um one thing I would add is I think um
another aspect to this this part of the
framework wherein you have this value of
prediction error kind of epistemic and
affective emotional value of prediction
error it's one of the kind of key
elements in this notion of uh agency get
in active inference that I think is
really it's gonna be really important in
mind as you go through the further weeks

of this course as well so what
you know like I said in the lecture it's
not necessarily speaking to a quite the
metaphysical question of free will but
it's certainly moving us away from this
picture of of the agent as a kind of
automaton that's just following rules
There's real space here for kind of
individuality and creativity as well
okay Darius has been waiting a little
while so
um let's go over to him happily patience
happily
um yeah I mean this is this question I
guess is deriving from my thoughts about
the meeting of the the high road and the
low road
some of the recent work that's been
coming out on Maxwell ramseed an actual
conversation I had with him which is
that the generative model of the states
so my question given that
give it an tall
is regarding the notion of affordances I
think it's natural that as cognitive
scientists and philosophers we take the
perspective intuitively of the agent in
the agent to reading a relationship and
how the agent is in the business of
reducing prediction error I was just
wondering whether
as philosophers
um you and Mark have thought about what
it would what is it like
out for the whole system to be in the
business of producing its uh prediction
error so I'm interested in flow States
um and so for example my thinking is is

that in the canonical example of
Thursday let's say a rock climber the
the climbing ball is also in the
business of losing its prediction error
as to afford the opportunity for it to
be
exploited stealth or themselves are in
the business of reducing uh prediction
now so I'm wondering whether this
expansion of affordances is
bi-directional traditional infinitely
directions better
that's an amazing request so um you
uh the Revenge the Axel constant that
somebody uh someone else but I spoke to
Maxwell ramstead last week
um yeah yeah
there's a there's a paper on uh I'll
find it in a second on uh Nishka
instruction
um and
um affordances I think it's
the Symmetry between uh a niche and an
agent
um very much in like the same vein as
you were just talking about there this
idea that it's not just uh there's not
just this unidirectional kind of fit
between the Asian and the environment
but the even or the niche is actually
modeling generative model of the agent
as well implicitly um
so I think the best I can do that
direction of some really interesting
work on that but it's um certainly
it's not it's it's not something that
plays a central role in my thinking
recently

um yeah so we have the participants
Zoom chat
yeah
a variational approach to Niche
construction yeah I don't know if uh if
Mark has anything to add or anybody else
um
which we're thinking yeah
yeah can I just check one thing you
asked I I don't know how much I need to
add here but um did you think that the
wall was reducing free energy relative
to the climber did you say that that's
right so my
so I don't know how that can that be the
case can that be the case the wall the
wall is a thing is maintaining itself
and I mean relative brain self-evidence
thing right yeah but but it's not really
a duet of one here it's not like it has
a generative model that's deep where
it's modeling the climber modeling it
like the same way as tango dancers do
the Tango the a Tango Duo are are are
having Cooperative flow States because
each movement opens a Vista of possible
error minimizing opportunities for the
partner who in turn opens of this set of
error minimizing opportunities for the
partner yeah and backwards and forwards
into this ever opening expanse of
affordance capabilities but an inanimate
object in an agent don't really have
that same Dynamic so I was wondering
what you were thinking there so my
thinking was is that this kind of bleeds
into the notion of fluidity of
affordances in the kind of gibsonian

sense and the fluidity of Concepts so I
agree in this in the sense of physical
self-evidencing it needs to model the
rain it needs to model the physical
environment so as to maintain itself but
what about as it's a it's a it's
because
so a sheer rock face is not a climbing
wall and that's because it doesn't
self-evidence as a climbing wall it
doesn't offer the affordances to be
climbed that's why I was kind of
referring to which is that affordances
change based on the on the agent Arena
relationship right your cause
affordances change whether it's working
or broken down similarly the climbing
will changes whether it's got footholds
handholds or whether it's just a sheer
rock face that was my thinking it seems
slightly spooky it seems slightly spooky
to me that we'd think of the Wireless
self-evidising in that way I think I
think the language that we tend to use
because Julian kivocene Eric reidfeld
um you know the Bloomberg and folks
often use the idea of affordances and
Fields of affordances in the active
inference way
um the the radical end of the pool there
is typically that the affordances don't
only happen on the agent side but rather
happen from a dynamic of the changing
volatile environment and the generative
model of the agent and that their their
collaborating in Dynamic ongoing ways
such that the affordances are emergent
between them that's certainly that

certainly seems right to me the the idea
to push it a little bit further and
think about the wall self-evidencing
it's a it's a I think it's a step too
spooky for me
um I think I would I would feel safer
safer a little bit closer to thinking
about yeah the affordances are changing
relative to the Dynamics of the wall
um but I don't know what it would mean
for the wall to be evidencing
the climber or itself yeah there's uh
just to kind of that's instead a ton of
thoughts well
a little longer on um
affordances because there's this really
nice distinction in their work between
uh what they call well I'm not sure if
they I think I think that the term
landscape of affordances is a little bit
older but they make this distinction
between a landscape of affordances and a
field of affordances
and I think the reason this is it's
really important to get on the table is
because it this distinction is kind of
directly targeting the dynamic shifting
very affective nature of affordances so
on the one hand you have this landscape
of affordances that is relatively static
so so right now my landscape of
affordances is is Brighton
um and that's the landscape of
affordances in that sense it's not
really going to change very rapidly
unless I kind of jump on a plane and go
somewhere else but the field of
affordances has this thoroughly kind of

normative affective character to it so
depending on my internal State at any
given time depending on my expectations
or my desires say my field of
affordances is going to shift very very
rapidly and it's so not just not just
predicated on my internal state but also
kind of contextual cues as well so if
something were to happen in the
environment that was afforded very high
Precision waiting then that's likely to
shift my feel affordances significantly
um
I think that the reason that I'm kind of
emphasizing this to darius's point is
and and kind of building on what Mark
said it it doesn't seem like this
distinction at least would apply from
the perspective of the perspective of
the wall say because this kind of
affectivity and normativity just isn't a
feature of the world's experience of the
world and also to kind of run with that
the I mean affordances perhaps you could
say a bit more about how you would kind
of think about this Darius because
affordances are kind of opportunities
for actions okay so it's they're they're
kind of uh I I like that there's some
debate about this but I like to think of
them as like a relational property
between of like this you have the
skilled capabilities of an embodied
agent and then you have some feature of
the environment so I think when you're
thinking in from the perspective of the
wall you have the features of the
environment because you have the

embodied characteristics of the agent
are kind of playing that role but it's
not clear to me what the the kind of
skilled capabilities of the wall is it
so yeah if you could just say a bit more
about that yeah I mean it may be the
case as you pointed out that actually
the the technical term affordance as
being obtuse faction has misplaced her
okay so I can see I again this is a lot
of this I've been formulating having
read this very new paper by um Ramsey
and colleagues on Bayesian mechanics of
physics of them by beliefs where he
really Rams home this point that the
system the generative model is the
system of the kind of statistic uh
stochastic differential equations across
the state space so across the mark of
blanca across the particle with its
Market blanket and the external States
and that is the system which is in that
system itself is in the business of
reducing free energy so it's not just
the active inference agent yeah the
whole system yeah which is embedded in
the whole system of other things and so
that was my general thinking is that
yeah just on the prerequisite that these
systems exist just on that aprily Axiom
there has to be some kind of self
evidencing of that system itself so
maybe affordances is the wrong word I
yeah yeah that might be the trick right
there as soon as you provoke affordance
yeah you're provoking phenomenology and
so now we're not talking just about
statistical variations now we're talking

about lived experience that might be
that might be the the linchpin right
there I think I think that is right yeah
and I and I would say I think Darius
you're absolutely on the right track and
you're in I mean there is this ambiguity
about affordances but aside from that I
think you're absolutely right I mean
it's
I it's just nonsensical to think about
an agent's absent an environment under
active inference you have to it it is
this agent environment system in its
totality that's minimizing free energy
they have to be they have to be taken
together as one and I know Abel you've
had your hand up for a little while
though do you want to is there something
you wanted to add
yep uh do you hear me well we do I do
anyway okay
so
um about the references World Arena
foreign season
um basically the symmetry of hydroxine
environment is a property of active in
France well of the financial principle
so if you buy a friendly principle and
you also buy that it entails that agent
have affordensies or any proximal notion
then you also buy that rolls as uh
forenses
so there are basically three position
you can have on that
one is that active inference is correct
well a [__] affordances and they do
self-villiancing which I would not go
with

another uh option is that you have
basically um Devol in the details so for
example Maybe
the adjunct environment is not the wall
it's something that is broader like Gaia
system whatever and it just happens it
includes the wall but it is not the wall
and it does do self-balancing
and the third is that active inference
is wrong uh we don't have affordances
based on whatever is the follow
presentation of active in France
and I would go in that direction
of this phenology phenomenology stuff so
we have a
I I would say philosophical evidence in
the sense that they are denying this
would lead to nonsense but the
information is based on observation
and some system do observing much more
actively than others and maybe it's a
good thing to have in your formalism I'd
say uh so um you have maybe a stronger
case for uh the way uh if
are constructed in the quantum
formulation of the fep if there is such
a thing which like Chris Fields someone
who we should Baseline believe claims
there is
because then you have formalization
which is not in terms of Dynamics per C
so causal constraints but observation
and indirectly phenomenology maybe you
have something that is stronger there
but
right now the uh link oh and yeah sorry
there is actually a bunch of people who
have a quite good hold on science which

are the
SE
based on the ecological activity and
commission on one hand and between
cognition and the metabolism the
constitution of the living thing that
cognites on the other
and one of their
definition of agency cognition pick one
entails
um
interactional asymmetry I think is the
word
that maybe something we would need to
translate into active inference well
into the fep for active inference to
have a strong grip over this affordance
thing until then we have a kind of hand
waving way it relates it to
more conceptual approaches uh that are
related to an active and ecological
approach to psychology and um
yeah the formalism lags basically behind
the concept for now
sorry for the long talk
no that was great much appreciated
this um
anybody else want to come in on uh
question of affordances because I know
um affordance has played a fairly
substantial role in the lecture so
anybody
the questions related to affordances or
you know action perception affordances
then Now's the Time
yeah I do have one but it's not well
structured yeah
I've been thinking about for the last

months because I I started to deepen the
artificial intelligence field up coming
from a Humanities background and
philosophical background so I'm mixing
many things in this period
mainly following the the active
inference as a as a Crosser between many
disciplines that and this is very
beautiful about the framework and I was
super curious about the um the question
proposed by Darius and I was then
thinking about
now we're actually facing a
a new environment and a new landscape in
which we're like interacting with the
term
comes the next turn actually generating
over revolutionary time and
synchronizing
a logical alien agencies operating
within our
landscape and field of performances
um so I think there's
a lot of really interesting work to
potentially be done there I will take
the opportunity to shamelessly plug a
pre-print that Mark and I have released
recently where um we one thing so one
thing that I'm really interested in that
I can kind of speak with some confidence
on is the fact that I I do think there's
a point at which the theory of
affordances or the language of
affordances starts to break down
legit
so the preprint that I'll show around
is looking at and um ambient technology
specifically so this is technology that

you the whole kind of impetus behind its
design and conceptualization is that the
user doesn't have to actually do
anything so it precedes the user
pragmatically and epistemically it knows
what kinds of things you want it to do
and it just does them in the background
and it works by shifting kind of it it
subtly shifts the material environment
such that it impacts your field of
affordances in real time
and and the argument that we make in
that paper is essentially that the
affordancers approach doesn't really
work for this and that we need to kind
of think again so
um yeah I mean I think that that might
be it but like
how to conceptualize generative AI for
example under the affordances framework
I'm not sure I don't know anybody that's
done any work on that but it would be it
would certainly make a really cool
project
you are making it better and right now
when was that
you are making it right now I know
because you sent me the paper to review
oh yeah
thank you thank you so um we were
looking at we so we uh
or early stages of a a paper on the role
of um generative AI in classrooms
um so from the perspective of
thinking about what kinds of affordances
generative AI represent in a in a
learning environment
um particularly we're kind of applying

active inference to classroom design and
looking at it from the different
different uh the angles of different
um educational and pedagogical theories
um but it's um
very early today with us
yeah this is I suppose it probably
depends on your view of generative in
general I mean it depends whether you
would consider it a real cognitive agent
or not
I think I think some people are more
inclined to think of it as you know this
thing this is a thing that has agency it
has real understanding it has like
um you know and other people are maybe
less inclined to think of it in those
terms and I think that might be that
might bear significantly on on how you
think of it
yeah that's super cool that basically
the last project you mentioned is very
very in line with my PhD project right
now because my main PhD project is our
European findings and it's about the AI
for personalized education and I'm
trying to follow it through the the
active in French framework from a
multi-level perspective
so yeah very cool yeah we should we
should probably talk more about that
yeah
okay uh Darius
yeah I wanted to ask I mean this may be
directed more marks I know this is kind
of his work in terms of
um slopes of uncertainty and about doing
better than expected at reducing

prediction error over time
I was wondering kind of how the
architecture of that is built into the
into the particle into the into the
genitive model because we have the kind
of for me there's I don't know if it's
an hour and out sort of conflict but we
have these kind of implicit priors that
we're going to fulfill certain
expectations or minimize prediction now
regarding certain things right so
homeostatic priors or happiness
well-being whatever it is but then your
claim is that we have a higher order
beliefs the higher order predictions
that over and above that I also need to
become I also need to do well better
than expected a reducing prediction
error over time so is there a it's still
quite fuzzy in my mind but is there a
kind of potential conflict there between
the general priors that the system has
and then go actually in a sense
violating those Expectations by going
over and above them which itself
constitutes an expectation how do you
kind of resolve that tension
yeah that's interesting
um but this is very much in Ben's
wheelhouse too
um Ben and I have been working on slopey
stuff for uh almost as long as I've been
doing
um
um
active charge
what was his original paper called we'll
get it we'll get it up they do a really

good job of showing technically where
this sits within
um uh the parametric model
um Laura senved Smith's paper on
metacognition also does this by showing
you have these you have these depths of
modeling where models above are modeling
models below okay and optimizing over
those models below so we have some good
computational backbone for thinking
um for not only thinking that this is
the case but
beginning to express how it does the
work it does
um so let's let's leave that for digging
into it though technically on sort of on
our own let me say sort of a at a higher
more abstract level still hopefully it's
useful
um it's not weird to think that this
kind of anticipatory system is not only
making predictions about the world but
it's also
um part of what it's predicting is how
fast or slow how efficient it is in
particular contexts at resolving certain
kinds of errors that that's a perfectly
fine thing to think that we're also
predicting slopes of Engagement
and then and then all we're saying then
is system also pays attention to when
those expectations are breached it's
learning from those breaches that's that
should just be the bread and butter for
what the system does anyway so Precision
is a second order is a second order
process in much the same way Precision
is about how well uh how how reliable

are lower level predictions
um and then using that
um that amount to toggle how impactful
either errors or predictions are so
we've already got baked into the system
right from early days this idea that the
system is not only making predictions
but monitoring its own predictive
processing regimes and then toggling uh
based on how reliable those those
substreams are
um all we're adding here is
um you know when we first when we first
thought about those mechanisms one thing
that can happen this is just good for
everybody who's interested in active
inference because I bump into this with
my students all the time
when we say something like Precision
waiting
um
a tendency to think it's the same
so we go to find what is precision like
where's the biological instead or I
guess Precision waiting when we actually
get into a bio system and we look for
these things the truth is precision is
going to be weighted in lots of
different ways I mean that's the real
Frontier of This research is to actually
find how these things are instantiated
and the answer is going to be
multifarious I mean it's going to be
you're going to have Precision
adjustments happening throughout the
system in lots of ways it could be
synchrony and asynchkenies and
desynchranies between systems it can be

neuromodulatory chemicals it can be
structural structural
um uh shapes within the brain I mean
Precision is going to be set in in lots
of different ways so
um
uh uh all all that we're pointing out
here is is that one of the ways that
Precision is being set one of the ways
that the system is tracking how
efficient it is and then upping or
lowering the amount of impact
um error signals or predictions have is
that it's happening in an embodied way
so we've looked over to the effect
Active search and and lo and behold
there are all these signatures that we
were looking for for this kind of for
this part of the Machinery so
um yeah so not weird that the system is
tracking its own regularities and
adjusting that was baked in right from
the beginning
how it does that
that's one of the Frontiers it's going
to happen in lots of different ways lo
and behold affect the Dynamics the shoe
fits they look like they do that stuff
and then you bring it to the lab and you
go back to like reward prediction error
research and sure enough neuromodulatory
chemicals reward systems are tuning
affective Dynamics relative to better
than and worse than slopes of
uncertainty management that's exactly
what we would expect given the
computational model
um so then it's an easy it was an easy

next step to start saying well look
there's one of the way
it's a Precision that there's one of the
ways that after
effects
[Laughter]
system now just notice there
and now the only thing the affect the
system is doing we never want to say we
never sort of we're careful not to be
reductionist to say oh affect is always
aerodynamics I don't think that's right
I think it's that aerodynamics are
expressed in part effectively and they
have this impact on the system
um that we can that we that we've known
about for a long time even from just a
reward prediction error um literature
does that help or was that a bit is that
okay
yeah yeah yeah uh yeah uh just to just
to add two two things really quickly
there um because I know time is kind of
catching up with this uh I would
emphasize as well like just as a kind of
General point the one of the things I
really like about the work on
aerodynamics and affect is it has this
nice kind of broad capture of the kinds
of affectivity that agents can
experience so we're not just when we
talk about affectivity we're not just
talking about full-blooded emotions or
even just moods but um we're talking
about what is it one of the things I
kind of Drew attention to in the lecture
was Matthew ratcliffe's work on
existential feelings which I think is

just such a um that the connection
between active inference and
phenomenology that you find in some of
this work is just insanely powerful and
it's one of the things that really drew
me into the framework
um the second thing just build
um
one thing that I don't think I mentioned
in the lecture is uh active inferences
it has been described as a
quintessentially metacognitive framework
so you have kind of built into the
architecture you have expectations over
expected so you have kind of predictions
about predictions and this has proved to
be
um just you know again just phenomenally
useful for thinking about certain
aspects of phenomenology as well so
um yeah I think that these are these are
like real strengths of the framework
okay
um I don't know how we're doing for time
are we are we strictly limited on or can
we we are not as far as I know
Danielle
well
um I can probably do I'm definitely do
another 15
um if anybody has any more questions
and people I should say as well we're
not that doesn't mean that we're all
held captive
um to my time frame so if people if
people do need to leave within the next
five or ten minutes that's absolutely
fine but um I'm certainly happy to to

carry on if anybody has any more
questions or comments
um if if no one else has got any I would
quite like to go back to uh earlier in
the discussion you were talking about
the value of uncertainty
um because uh
in organizations in which I'm working
with
uh we often have conversations about
uncertainty and what we find is over
time there tends to be a drift towards
risk aversion so we're working with a
large uh International construction
company at the minute and that they used
to have a culture in which innovation
um uh was uh was quite well embedded and
and over time it's kind of drifted
towards very very risk-averse culture
where actually rather than learning
through the process and being able to
tolerate uncertainty you know around
opportunities and you know what can be
accomplished and that kind of the
valence that comes with uh the intrinsic
reward I suppose of being able to reduce
uncertainty about capabilities to a
situation where they're trying to
anticipate
all of the risks up front before they
even get involved in the project so you
know there's some kind of valence switch
market and you mentioned um David Blaine
earlier you know and something about the
context you know there was a switch in
valence from yeah this is this is a kind
of a play this is a safe thing to do to
actually know this is you know this has

real consequences so I'm just kind of
I'd like to explore that a little bit
more actually if anyone's got kind of
any insights or papers or comments
yeah so I think that's a really
interesting question actually and I I
think um
so it seems like
one of the things one of the things
we've been talking about is this
connection between uncertainty
prediction over minimization and
affectivity and individual agents and it
sounds like what you're asking is like
how does that translate to uh like
collectives so yeah one of the things
about active inference that we're going
to see in subsequent weeks is how it
scales up to kind of larger systems like
companies or you know groups that are
trying to achieve some kind of shared
goal
um how do you how does the value of
uncertainty translate onto like how is
it scalable in that sense right is that
does that be a fair kind of
yeah I I guess so
um I I guess it's um
I mean I
so uncertainty often when it's talked
about uncertainties talked about as risk
which is you know I suppose is you know
predictions or anticipation of uh
outcomes that we don't want right
whereas there's also positive
uncertainty right you know which is you
know I suppose you know simply stated
opportunities you know through curiosity

you know what might we be able to
achieve you know this kind of novelty
seeking I guess that you know new
opportunities that we haven't exploited
um and they're just you know in the kind
of organizational systems uh you know I
think there are various pressures that
cause it but but over time you see these
cultural shifts towards uh you know a
very very kind of risk-averse you know
where you know where the valence is
obviously quite negative you know you
know quite an unpleasant feeling for
people and yeah yeah
this is good this this is right at the
heart of my current work
um I'm really interested so you know we
just had a big stint where active
inference models were starting to be
used in computational Psychiatry
especially for pathological disorders so
addiction depression disassociative
disorders OCD PTSD
um it's a really it's quite a it's a
quite um a sexy framework for thinking
about some of the ways the cognitive
system breaks down in the sort of new
move right now I mean we have a
collection coming out right now in
Neuroscience of Consciousness is to
think about these things in terms of
okay
if we are predictive systems and we have
a sufficiently rich model of how that
system works in a particular Niche what
sorts of what sorts of um what sorts of
ways can we intervene on that system in
order to have positive outcomes rather

than just modeling what the negative
outcomes are and I hear that a lot in
what you're saying now so I'm just going
to drop one link for you there this is
Casper hasp again
um wrote a little a very small paper
with a nice little model called
sophisticated affective inference gives
it a little bot that tends towards
catastrophe
catastrophizing the future if it's given
if it's given like fun medium fun medium
fun dangerous over time uh it tends to
it tends to expect the dangerous one
um uh like after 10 000 iterations it
basically lives in the worst possible
scenario which is a nice little this is
my wife
this is this is most people today you
know so the question is um I want to
know given them all model why does that
happen and what can we be doing to
intervene and um part of the answer is
going to be we need to become more
tolerant of uncertainty that's one of
the things that the system can be better
or worse at so this just takes the
computational modeling and then looks to
things that we already know about
emotional regulation that that's that's
part of the story
um so
um we've done a little bit of work on
this with our predictive dynamics of
happiness and well-being and Ryan Smith
is definitely doing work on this with
active inference and well-being
um so definitely check that out if

you're if you're interested here
um
I'll just flag one interesting thing
here that relates just to what we were
just talking about about layers of
modeling
one way you gotta let's say two ways
there's two ways that a system and it's
probably gonna be lots but the two that
come to mind that A system can become
more tolerant uncertainty is one
exposure
so this is why exposure therapy might be
useful for our kind of a system you want
to expose the system to volatility at
the lower and middle levels of the
hierarchy and have it turn out okay so
that this one of the things that the
system can come to predict is not only
you know particular outcomes but it can
also predict how much error is involved
in particular outcomes so for instance
when I used to give a pro talk I was
always really nervous
um and I don't know if that ever really
went away but I've had so much exposure
to the anxiety of giving a professional
talk that basically I don't notice it
anymore but if you were to ask me at the
beginning of my talk
Mark right now what's your phenomenology
and I sort of meditatively looked I'd
probably say yeah I'm nervous but if you
hadn't have said that and you was like
hey what's up I've been like oh yeah I'm
great like it's all good I don't even
notice it anymore that's because the
system knows that I have errors

but as soon as I start talking they drop away and because I know the arc that evens error in the system it becomes non-newsworthy it's no longer interesting volatility to be tracking that's just from exposure okay so what's happening is you're having errors at a lower middle level that a higher level is now modeling saying when we come here we should expect this Arc of error same thing when you work out like real gym routes they can feel good the legal system right but at the higher level it's now learned not only that that has a natural Arc but that that's a good sign at a higher level so now you're getting a positive you're getting positive prediction error slope high and negative prediction air slope low okay so one is exposure two you can model your own responses this is something Ben and I work on with um horror movies um uh you can you can you can take an active role in mindfully observing your own reactions to volatility and um this comes up in our paper on horror we've already dropped the link today if you want to check it out something really special happens when the system models its own reaction to volatility it starts to learn that um even reactions to volatility uh don't need to be um they don't need to be compounded up into dangers that it's okay it's okay to be uncertain at certain levels of the system so how that translates into business I'm not sure other than I know tolerance to uncertainty is a marker of

business success and
um I suspect the framework in a fit lots
of mindfulness mindfulness work about
learning to tolerate uncertainty
is there anything about
um the role of language in kind of
modulating those metacognitive oh it's
good if I'm I Was preparing an answer on
that specifically and I was nervously
waiting for an opportunity
so
um
like meta in terms of evolutionary
cognitive archeology we don't know when
language emerged but we do know that's
uh let's say an encouraging semantic
system so
you can build things in language and you
can expect things in the world to
correspond to things in language
so for example if you that's abstract as
I say it's abstract but you can for
example have a self-identity
French I'm a French
and I communicate with the expectation
that are underneath with other people
and that means we're communicate our
nose so so we have specific expectation
of what we will
do and those expectations they become
embedded in specific symbolic markers
that are embedded in language
and uh something that uh like this thing
that it was strongly evoked in my mind
when you talked of three words to answer
that is that um
as far as documented history goes
uh Europeans well indo-europeans they

are pretty strong on the idea that
things of a nature and they act lawfully
and their nature and their laws they
somehow correspond to linguistic
categories so I can say stuff like
chickens quark and that's a proper
explanation of what are chicken and why
they quack
and if you look in Chinese philosophy as
far as written history goes you have a
major Accent on
way way so
um a poor translation would be uh
effortless action
closer translation would be uh action
without action
so it's a form it's something that is
quite close to the phenomenology of flow
in that you observe yourself doing
things and you do not apply a conscious
effort to the flow of what you're doing
and that looks like something that is uh
closer to the phenomenology would expect
if you apply you know recursive
proactive condition with less or less
heavy or more effectively integrated
um
symbolic occurring like use of language
as something that you actually expect to
be meaningful and to constraints from
your actions
and marginally I'd expect very strongly
do you expect that linguistic categories
will map and what you build with it so I
don't know self-identities plans
whatever will map cleanly onto What You
observe you will be very very unsure
about things because that does not

happen usually
um
and yeah um besides uh Grant discourse
on interpreter preference traits which
are usually not that constructive
I'd
um
yeah sorry I got lost because I did not
I that's a speech tonight can I let me
just say one thing before before we move
off this point uh a simple way that
language might help is by invoking
cognitive flexibility so
um oftentimes we think about the
management of error being either
updating predictions or acting on the
world those are the that's the dyad
usually what it overlooks is the third
one which is we can also manage
volatility by redeploying Precision in a
better way
so rather than just updating your model
to fit the world or updating the world
to fit your model you can just change
the set of what matters so that you're
not you're not really updating the model
and you're not really changing the world
you're just changing the problem
landscape and that's something language
allows us to do it's maybe one of the
one of the really amazing things that
language allows us to do is we can use
language to bootstrap that kind of
precision adjustment so if the train
doesn't come on time we both notice it
okay and we feel bad that's perceptual
updating we might go and get a taxi
that's active updating but I might just

say to you
um wow look isn't that isn't it lovely
that now we get a little bit more time
to read our book or to continue our
conversation or let's finish our coffee
with a little bit of ease I mean we get
an extra 20 minutes now even just saying
that redeploys Precision over the
problem space now the train being late
isn't volatility in the system
the train being late is signaling to the
system that a better than expected slope
has been achieved isn't that so
interesting the exact same occurrence is
either undigestible volatility right
like you're gonna be late oh my goodness
or it's an opportunity for an
improvement in the system which is now
you get more time with the person that
you're um taking the train with and that
was just a matter of linguistic um
perturbance of the way that Precision is
being deployed so I if you're looking at
if you wanted to dig into the research
here I would look up cognitive
flexibility and language or coaching
cognitive flexibility is such a good
point here
um
so the uh you have a paper by any clerk
on uh this specific and current role of
language and something it entails is
that basically you got like if the
active influence picture is correct you
will flexibly predict to flow in the uh
interceptive proprioceptive
exterosceptivist not even a thing space
and that is that is what Builder

integrating experience integrated you
know cognition
and language it adds another layer of
complexity if you can just talk to
yourself in your head that is another
dimension that you can predict and that
can be coherent or incoherent with uh
you won't
and so that is an extremely powerful and
current system because I just have to
like tell a plant to myself and boom
that uh pushes me nudges me nicely for
the plan but then you can have a
different level of meta expectation over
how language corresponds to reality and
I'd say a very very strong factor of you
know a version of coincidentally is
whether you expect your plans or your
linguistic structure to nicely map them
to reality because it will not happen
but if you expect strongly that it will
be the case you will have to make it
happen somehow and here you will adapt
rigid strategies and you will
like lose flexibility but also language
and sorry for flexibility to be the case
in the first place so it's about
coupling between dimension of cognition
yeah and I would just uh just kind of
this
risk and danger are two things that I've
been really interested in across a kind
of variety of different contexts and I I
would say
some of the things that have been said
like the kind of how you can
influence the system by externalizing
things through language and the

different strategies that systems
whatever scale they're on have
minimizing prediction error
um minimizing free energy but also I
would just emphasize the importance of
like external context here as well so
even in the case of like a collective
like a business you still have to take
into account the environment in which
the business is operating and I I've
been really interested in a particularly
extreme example of context
um I used to do still do some work in
philosophy of sport and I'd be really
interested in dangerous sports and why
there seems to be this insane contextual
effect where within a very kind of
narrow contextual band of sporting
practice people seem willing to take on
risks that outside of that context would
just be completely insane
um and I think this is something that's
probably going to come up again again in
uh you know subsequent weeks as well
here when we talk about
um the way that expectations on an
individual or a collective basis can be
set but through other Minds uh like
patient so what somebody else expect
myself to do and and so on and so on
uh Darius you've had your hand up a
while do you want to jump in yes please
I'm not sure this is the sort of grand
synthesis of any of this but um
hello was mentioned as I said that's the
kind of
um
at topic of mine and something that we

know about flow and this way it's
integrated into this idea of language
and agency
is that at least in its original
conception of success mahali and the
sort of qualitative experience
associated with flow you would find
stuff like not only the dilation of time
but also the reduction in
self-consciousness
and now this is sort of picking up a lot
of work that happens at UCL and people
like Jeremy Skipper how integrated
languages into the sense of self
and it seems to me and I you know this
kind of shooting from the hip but all of
these very deep-rooted I guess they're
kind of psycho Technologies the self
language seem to dissipate at this
Goldilocks zone at this point of at the
at this edge of criticality at this Flow
State
which makes me think if the flow state
is a phenomenological offshoot of being
or you know optimally reducing
prediction error what is the kind of
role of the concept of the self or
linguistic functions because it seems to
me that that becomes a regulatory
function when that optimality is reduced
when soft starts going wrong
um so I think you can also think about
this in the kind of hydrogarian or
Dreyfus sense of just opening a door you
only start to represent the door in
yourself you only give it the linguistic
object of a door when you can't open it
when you open the door

standard there is no representation
going there I mean you could even argue
there's no quality out there so I'm
wondering how deep that runs and what
the kind of
what we can say about the role of agency
selfhood maybe even consciousness
um because they don't seem to be that
prevalent at least from my understanding
when we are at the edge of criticality
the the suggestion you're making there
is um
it sounds like what you're saying that
these
thing you call them psycho Technologies
which I really like uh you know like
language and uh and and self-mod like
certain aspects of self-modeling they
are kind of scaffolding uh or like a
ladder that you then kind of gets kicked
away once you reach a certain kind of
edge of criticality like where the
performance becomes
you're performing at such a level that
you just don't need those those
scaffoldings
yeah or or
the self-regulatory mechanisms that we
harness when we're not in flow are
linguistic or agentic in essence and
once yes once you saw you you don't need
that when you're at the sort of yes when
you're at this edge of criticality
um and I you know I only postulate that
because I want I'm only I'm only
thinking how deep does that go because I
know obviously the the active inference
framework is trying to tackle in some

ways the hard problem as well
and there are some flotations of the
idea at least within the flow Community
about the kind of
I don't want to say elimination but the
moderation the modulation of qualia on
the flow States so could that could we
also consider that to be an a construct
um which happens when we're not
necessarily
optimally you know reducing prediction
error yeah so a couple of things I so
I'm certainly not the person to start
talking about Consciousness so I won't
um but I would just um
I one of some of the readings that I
mentioned in the lecture might be
interesting for you on this topic so
if we're talking about
certain structures that seem
very very pervasive in our phenomenology
and how those structures can sometimes
kind of dissolve away
and how that might be accounted for
under a active inference framework
there's some really interesting work on
psychedelics by George Dean and some of
George Dean's co-conspirators I know I
know Mark and Sam Wilkinson worked on a
paper with George so George Dean's done
some work on kind of ego dissolution and
certain kinds of experiences
related to selfhood
um kind of and I
kind of don't want to overstep the mark
either but I think they kind of come
they fall generally under this kind of
Rebus model of um psychedelic efficacy

so I don't know if people are familiar
with this but this kind of finds a
natural articulation through predictive
processing and hierarchical
self-modeling and and that kind of stuff
I also wanted to really quickly flag on
this topic I'll find the paper and I'll
put it up on the coder because I don't
think I'll be able to find it quickly
enough now but there was a paper that
came out very recently
um that showed the efficacy of
um internal self-talk on sports
performance
so there were so they did a they did an
experiment with cyclists and they found
that when you um when cyclists were
allowed to engage in kind of constant
self-talk with them so like in a
monologue in their head their their
performance at cycling was like
measurably boosted which I thought was
really interesting and kind of relevant
here so but it's it's interesting
because it kind of just kind of
uh it sounds like it might primaphasia
be in tension with what we were saying
about flow States as well right so it's
if if the real edge of performance is is
a place where these structures are
phenomenologic phenomenology bleed away
um it kind of calls into question like
to what extent are these psycho
Technologies effective and in what way
are they effective and
yeah there's like yeah just a whole kind
of universe really interesting questions
there

so if I may uh sell food indirectly will
be the the central topic of the last two
sessions
my position about it is that it's
selfful in a sense we use all of the
time
is quite high level of construct that
emerged from the ability we have to
compare our activity to a
socially embedded mother of our activity
so that is something that is quite
specific to humans because of their
linguistic ability and or because of
their tendency to um what tomasello
calls the share intentionalities for the
ability to collectively Define a neck
plants
and so basically this is something that
enables a very deep very um
profound and robust transmission of
cultural knowledge and transmission of
norms this is what enables the
Consortium of norms
uh
uh but it's quite heavy cognitively and
it is predictable that if you're busy
comparing what you do to what's another
idealized version of yourself is doing
you're gonna invest less Creative Energy
in actually doing the thing that you
would if you just you know did a thing
so I'd say you have a pretty direct
entitlement of the uh this flow thing
from this model
yeah yes this is my conclusion sorry
um I think so we've run uh almost 30
minutes over time and I know uh Mark's
marks had to go and I'm gonna have to go

very I think I pretty much have to go so
um I just wondered if anybody has any
final kind of quick questions or
comments maybe we can move towards
wrapping it up
um I'll I'll make a few comments but
then you you're welcome to leave and
other people welcome to stay
um if they would like
So yeah thank you
yeah yeah
well first um just on a
logistical note there was a little bit
of lag so I'll encourage everyone who's
listening as well as who's participating
that we eventually plan to as with all
of our live streams develop the
transcript which will be curated and
published and eventually not so far away
in the future that'll be enabled with
speech modeling to generate podcast
quality audio so the future will be
smooth we promise
um but a few really interesting pieces
that I that I picked out while I was
embodying and listening
um sir Regina you you opened the
discussion with the disc with uh
surprise and creativity and indeed it
sounds like it's going to be attention
how can a framework who's imperative is
bounded surprise as opposed to say
maximized reward how can a framework of
his imperative is surprise minimization
and bounding be used to generate novelty
and I felt like people provided a lot of
really cool answers and one other thing
it reminds me of is uh Doug hofstetter's

notion of spec fetishness I don't know
how do you pronounce it but s p h e x i
s h n e s s and it's the specs wasp and
so he says well forget this whole
creativity thing because when we talk
about creativity it's like some sort of
you know Pantheon like Divine status and
abuses have to be uh you know called in
and so then oh that's not real
creativity and this isn't real
creativity so instead he says well let's
just focus on what would be the opposite
and that would be the road procedural
following even when there's an
opportunity for what we would call
creativity and then there's a Continuum
so why does somebody paint that painting
well why didn't they invent a new genre
well why didn't they invent a new media
and so all activity is existing within
this continuum
of creativeness which has many features
such as generativity and productivity
but also novelty but abounded novelty
it's not creative to to knock the
chessboard pieces off in a way that
there might be an elegant chest move who
only somebody in a certain cognitive
setting could detect the Aesthetics of
but I think that that was awesome
um and then the second uh piece that
really was was powerful was what
Mark Mark's train the I guess the trains
don't run on time where Mark is but but
that's okay because he was able to
rather than treat that as an
undigestible surprise
that was able to be basically

cognitively metabolized into like an
opportunity for friendship
through language as a social medium
and then he connected that to cognitive
flexibility and coaching and then I
thought about self-talk and our inner
monologue inner voice
and then there is that that was very um
provocative that like we aren't having
self-talk potentially not even having a
self in the flow
manifolds the experience of a cell for
the speech self self talk or again
potentially even the total actual self
as a technology and so then like
healthy self-talk would guide us
to the flow
and be kind of like a self-limiting
technology
because it would be like used in order
to not be needed to be used
which is kind of how we would hope
Adaptive Technologies would be versus a
maladaptive technology would be
something that
um entrenches its own utilization
potentially at the expense of the
functionality kind of like Doom
scrolling status but that's internal
rumination and those are the kinds of
things that people have simulated in
active inference
so those were some really cool
pieces that was a great discussion
could I just uh just just add something
on the topic of creativity that I
should have said earlier as well because
I think it's really relevant to Regina's

question and just the topic of
creative and novel behaviors in general
and the right of inference I think that
whenever we talk about especially
artistic creativity there's a tendency
even even for people who are kind of
well very very well burrowed into active
inference and and activism and these
kinds of Frameworks from cognitive
science I think there's a tendency to
think of creativity as as still like
it's almost like the last Bastion of
internal cognitivist thinking like you
you think creativity is something that
happens in here and bursts outwards and
so how does that happen with this kind
of whatever cognitive architecture we're
positing but there's really very cool
work by uh have to give a shout out to
Mike wheeler
um who has written some papers on the
extended mind and creativity and one
thing that's not really come up very
much that I didn't get time to talk
about in the lecture is this really nice
natural marriage between the extended
mind and active inference um something
that Andy Clark has been working on and
I know he's gonna going to be doing a
lot more work on it but the claim
wheeler makes and some others um
I think Joanna zelinska has written and
written a little bit about this as well
is the fact that creativity itself is
just as subject to kind of uh material
in but the the the sculpting effects of
material environments and socio-cultural
environments as everything else is so

create creativity is not this kind of
romantic with a capital R kind of
internal process that kind of bursts
forth but it's creative thinking is just
as much extended and embedded as
everything else and there's some really
great examples in in Mike Wheeler's
essay
um I don't know is anybody familiar with
the band Alt J
there are British band they were kind of
really big a few years ago they won the
Mercury music award because they have a
super original sound
so Alt J have this really quiet tinkling
uh yeah they are a great band they're
one of my favorites and they they have
this really quiet pinky sound that
everybody kind of assumed was just the
result of some kind of internal genius
on the part of the band
and as it turns out Alt J originally
they they tried to rehearse as a as an
indie rock band but they were confined
to rehearsing in an apartment block and
they kept getting noise complaints and
so they ended up developing this quiet
tinkly sound because it was the only
sounds they could rehearse that wouldn't
get them kicked out of their apartment
and I think this is a really beautiful
example of how like even the most
creative processes that seem on the face
of it really creative are in fact
just a subject to these external kind of
this kind of agent environment system
that underpins the whole framework so
it's

that's something that's going to apply
to absolutely everything
awesome and just the last question
people are welcome to to drop off
um the next section that we're heading
into I'll be doing the lecture that's
going to be in August and it's going to
be on Collective Behavior so what would
people
like to learn or focus on about
Collective Behavior
anyone who hasn't um spoken or or it can
just be a thought question
but I haven't prepared the slides at all
so
I'm happy to take suggestions
yeah Darius and then anyone else
I think what might be interesting is if
we're saying that these uh systems are
all in the business of producing
prediction error of self-evidizing why
do we see a diversity of
um why do we see a diversity of social
cultures of norms of
um standards at a global scale why do we
not just see some
um
homogeneous way to reduce picture there
which would manifest as a kind of
singular culture which which is in the
kind of business of self-evviducting that
just kind of popped to my mind
cool
anyone else want to give a thought on
Collective Behavior or on any other
aspect otherwise it's been great
I would like to make a comment of
creativity

so but uh does anyone like it's romantic
if you want to make a comment on the
next session I don't want to put that
Ben said that creativity is
intrinsically very hard to model because
uh by definition
uh creativity is something that brings
about something new let us say
and the mathematics we have to describe
physics and life they are not very good
at new things
because the most basic tool you are the
basic way to represent a system that
literally everyone uses is a state space
which is uh the list of all
possibilities of a system
if you say something is creative you're
likely to think that it means it can
bring about new possibilities and this
intuition it conflicts directly
with the very basic use of math to
describe it so it is basically the same
issues I was referring to earlier when I
talked about the comparison between
active in France and an activist so
let's say uh
activity biology inspired model of
cognition and a core concept
historically in those circles is
autopoiesis so the ability of the living
to self-create literally this is Greek
for self-creation
and you are the a lot of drift and
conceptualization and ambiguities that
were resolved or not resolved and led to
the crazies of frame of blah blah blah
but today we talk of autonomy rather
than autopoiesis and with the final

autonomy as uh the property for a system
of constraints over the activity of a
system to reproduce genitals and because
we're talking constraints
uh just the ability to influence causary
outcomes
there is no really a prior that Dynamics
in this space would be conservative so
we have a possibility for constraints to
you know bring about new things and
reconfigure themselves and the notion of
hydrogen she had used is based on that
it's the ability to let's say
reconfigure constraints in your
environment
and always the notion of creativity
which would be likely a very close proxy
to that but it's something that we do
not know how to prevent
and most of the mathematics that exist
are structurally enabled represents and
so yeah that's a big one
foreign
Michael
uh I uh good to be here sorry I was late
uh what comes to mind for me at the
collective you know I mentioned units of
collective life what is it a wee is it a
we Affair and uh you know but you know
what what are the what are the units of
we and any thoughts or insights about
that uh the the stages of the emergence
of formation into a collective uh from a
me to a weed for example or um
uh because I had a one that was on the
tip of my tongue and it just escaped me
oh
um you know this idea of precision that

um that I think it was Mark Miller was
talking about or Matt
um saying um you know uh
at the uh
if there's actions
what might be some of the most well
understood limiting beliefs about when
we use language of collected that keeps
us repeating the same blind spots um uh
and how might we say differently about
Collective so that we we escape those
predispositions
make sense yeah
they'll do what I can do in the solo
lecture and I'll I'll look forward to
the conversation where we can unpack it
yeah
yeah exciting thank you for doing it I'm
looking looking forward to it yeah it's
awesome I'm Nathan or David you want to
add anything
all right then thank you all hope
everyone is um enjoying the course so
far thanks aval again for for
coordinating it and to Ben for this
great section now you can put on your
student hat again
so and three of organizing
thank you see you all next time
bye thank you cheers
thanks everyone